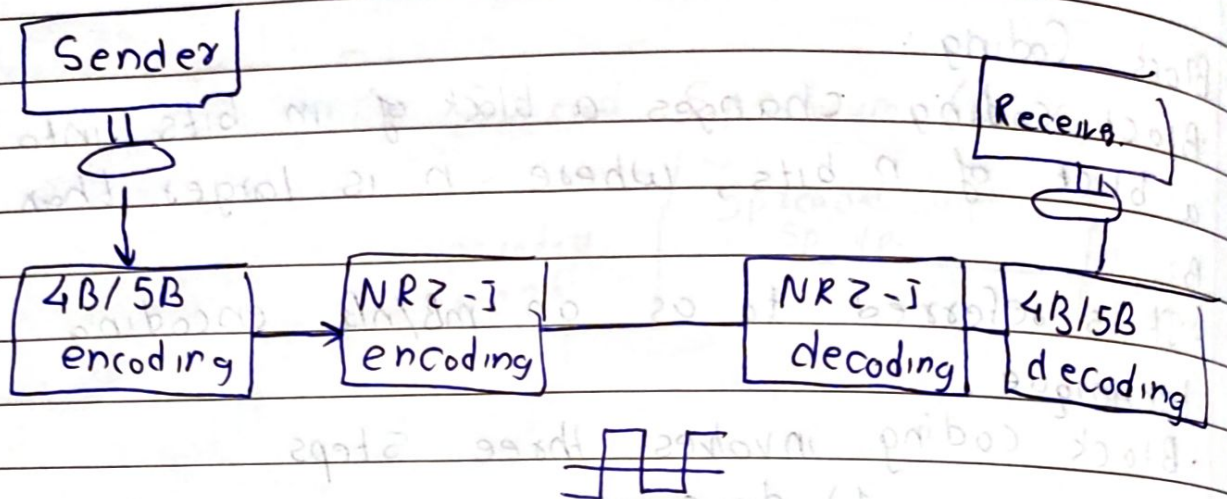




* Data Link Layer:

* Error detection and Error Correction

Media Access Control

framing

Ethernet 802.3

802.11 Wireless

Single bit error.

Burst error -

(Width of Noise Pulse)
↓
Data rate

→ determines how many bits get affected.

Error detection:

Error Correction:

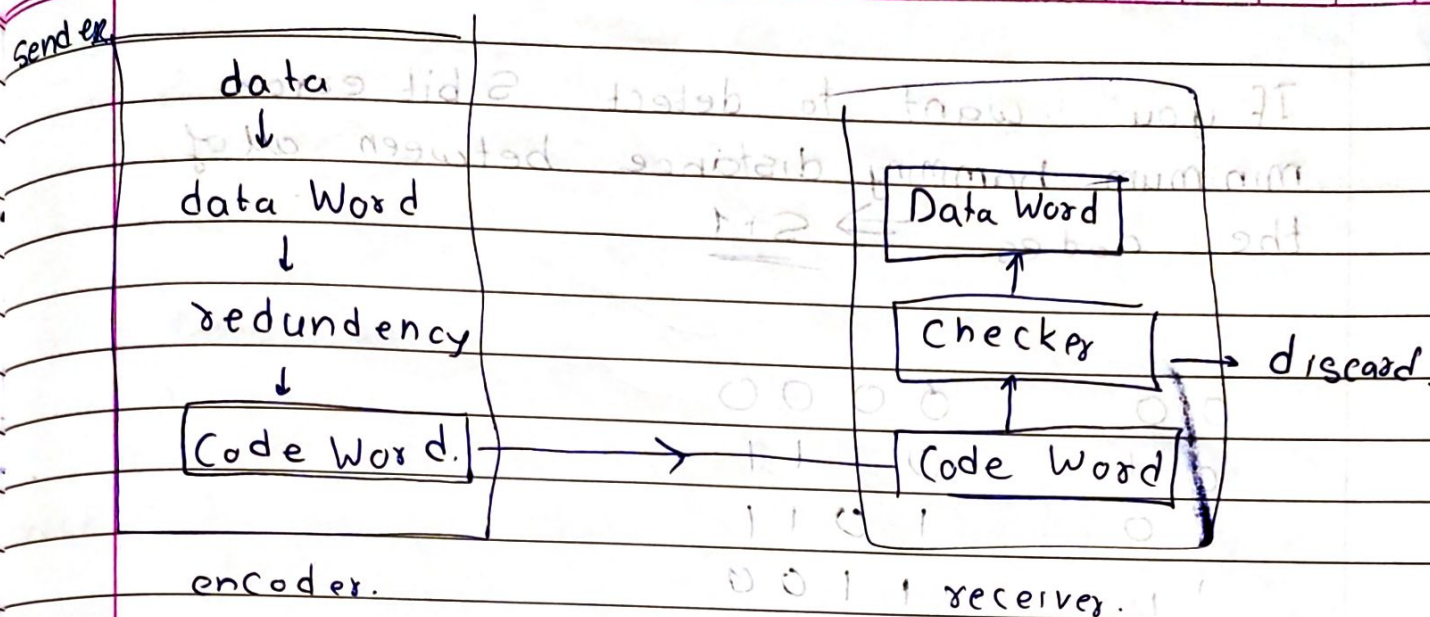
Some algorithm detect errors and some correct code.

$$m = k + r$$

Data Word → Code Words

* $2^m \rightarrow$ Code Words.

* 2^k - data Words



data Word.	Code Word.	Invalid Code Word
00	000	001
01	011	010
10	101	100
11	0110	111

Sender

Receiver

01 $\rightarrow$ 011 $\rightarrow$ 011 $\rightarrow$ 01

01 $\rightarrow$ 011 $\rightarrow$ 111 $\rightarrow$ discard

01 $\rightarrow$ 011 $\rightarrow$ 000 $\rightarrow$ 00

* Hamming distance:

Distance between two Code word after converting from data word.

101
000 } 2

101
101 } 0

$$d(1010, 0110) = 2$$

If you want to detect S bit error
 minimum hamming distance between all of
 the codes $\Rightarrow \underline{S+1}$

0 0	0 0 0 0
0 1	0 1 1 1
1 0	1 0 1 1
1 1	1 1 0 0

* Parity System: (Even Parity)

4 Bit data word.

$$R = (a_3 + a_2 + a_1 + a_0) \% 2 \Rightarrow 0$$

$$\Rightarrow 1$$

Data Word	Code Word
-----------	-----------

0 0 0 0	0 0 0 0 0
---------	-----------

0 0 0 1	0 0 0 1 1
---------	-----------

0 0 1 0	0 0 1 0 1
---------	-----------

0 0 1 1	0 0 1 1 0
---------	-----------

$$s_0 = b_3 + b_2 + b_1 + b_0 + q_0 \pmod{2}$$

$$a_3 \ a_2 \ a_1 \ a_0 \ r_0 \Rightarrow b_3 \ b_2 \ b_1 \ b_0 \ q_0$$

* Cyclic Redundancy Check (CRC)

data word $\rightarrow k$ - bits

When G_0 is 1 we discard

data word $\rightarrow k$ bits
Code Word $\rightarrow n$ bits.

CRC

Binary division

data word $\rightarrow 2^4 = 16$

Code word $\rightarrow n$ by 128

divisor $\rightarrow m - k + 1 = 7 \rightarrow 1011$

remainder $\rightarrow n - c = 7 - 4 = 3$

1001 001 01 01 01 11

```

      1010
    -----
1011 | 1001000
      1011
      ---
      00100
       0000
       ---
       1000
        1011
        ---
         0110
          0110
          ---
           110
    
```

1011 $\rightarrow$ 1001110

1001110

1000110
↑
error.

Cyclic Redundancy Check

1011 | 1000 ~~110~~

1001 $\xrightarrow{\hspace{2cm}}$ 1001110
Codeword.

Q. Find the codeword for the following dataword using divisor 1011

1. 1011 2. 1100 3. 0101 4. 1010
5. 1000 6. 0100 7. 1101

1011 | 1011 000

1011

0000

1110

1011 | 1100 000

1011

0110

01110

1011 | 01110

1011

00010

00000

00010

1100010

* CheckSum Technique.

Consider the following 32 bit (8 bit word)

1 0 1 1 0 0 1 1
 0 1 0 1 0 1 0
 0 1 1 0 0 0 1 0
 1 0 1 1 1 0 0
 0 0 0 1 0 1 0 1
 0 0 1 1 0 1 1 0
~~1 0 0 0 0 0 1 0~~
 1 0 0 0 0 1 1 0

Check Sum

1 0 0 0 1 0 0 0
 1 0 1 0 1 0 0 0

0 1 1 1 0 1 1 1 → Check Sum

1 0 1 1 0 0 1 0
 0 1 0 1 1 0 1 0

Calculate 16 bit

①	①	③	②	
C	2	4	9	
0	A	8	6	
2	1	E	7	
4	B	9	6	
1	0	D	8	
4	B	2	4	

20

4 B 2 5

→ 1's Complement

B 4 D A

→ Check Sum

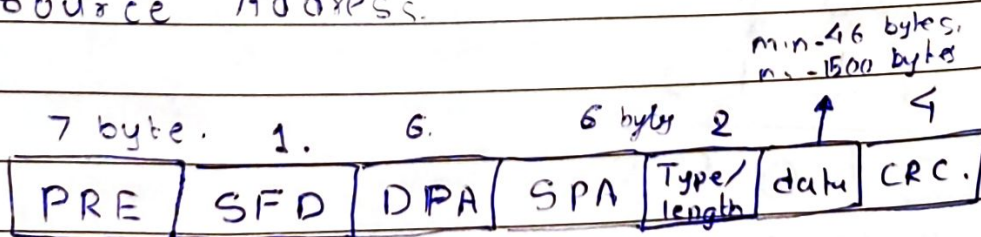
* Standard Ethernet :

Ethernet is a technology which moves packets from one machine to another.

destination Address.

Source Address.

Ethernet uses Manchester encoding.



10...

10101011

0xAB.

Start Frame delimiter
destination Physical Address
Source physical Address

CDPA - destination address can be unicast, multicast or broadcast.

• SPA is always unicast.

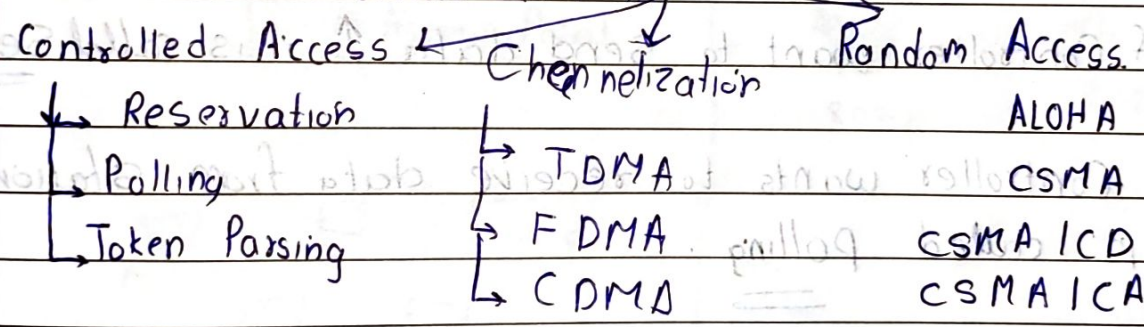
If DPA is all F. (FF.F) then the data is broadcasted.

The 1st of first octate (8) tells if message is unicast or multicast.

0 → unicast.

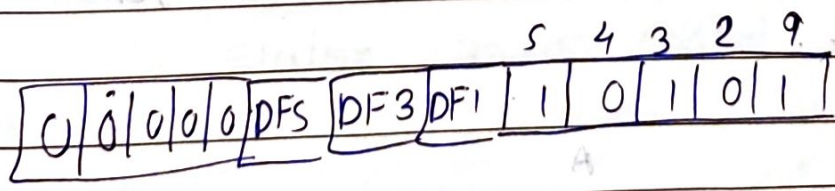
1 → Multicast.

* Medium Access Control (MAC)



* Reservation Algorithm

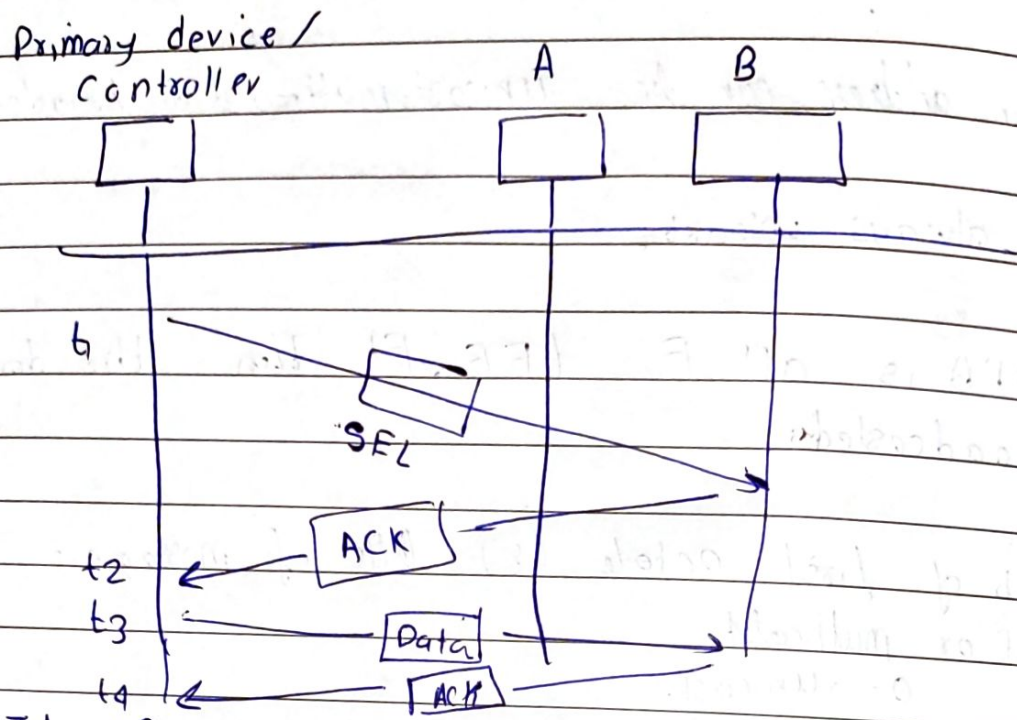
The is a reservation frame where Station makes reservation by setting a bit



* Polling:

1. Select Method :

Carrier Sense Medium Access

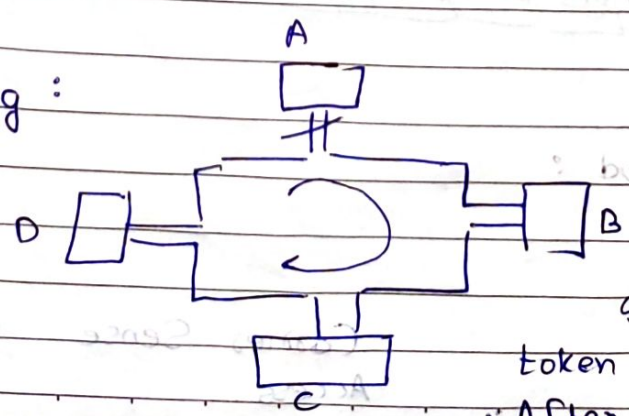


Identifying a station (Various Mechanism)

- If Controller want to send data $\uparrow$ it is called Select
- If Controller wants to receive data from stations it is called polling

1. Select a device
2. If device is ready it sends acknowledgment
3. Data is sent by primary device
4. If data is received acknowledgment is sent back.

* Token Ring :



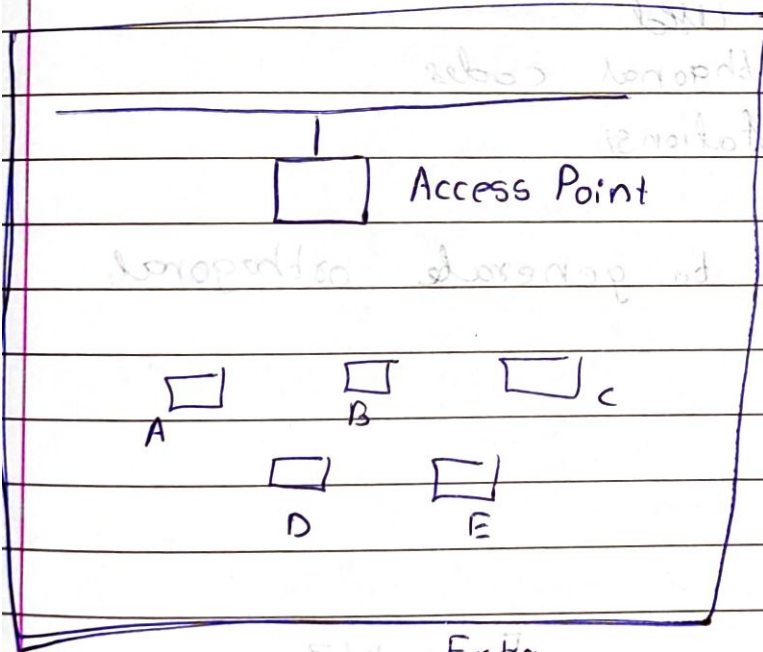
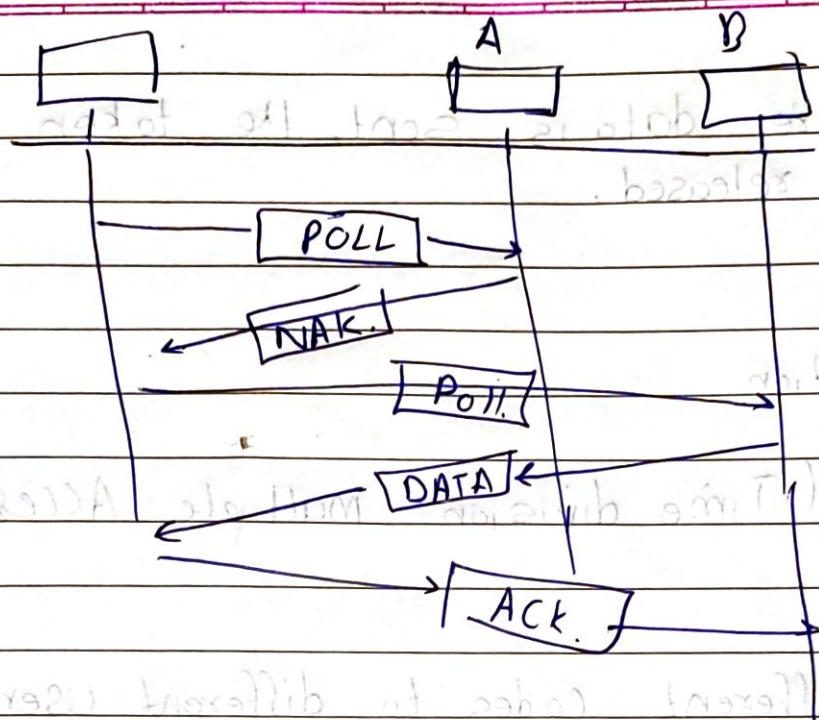
Token ring frame

is circulated

To send data a station acquires

token ring

After acquiring token



Wireless LAN

~~Access Point~~

ethernet

802.11

802.3

ring, the station gains control over the network.

Dest Add | SA | Data |

 Data Frame.

- It then circulates the data frame.
- After receiving the data acknowledgement is sent by recipient station.

- After entire data is sent, the token ring is released.

* Channelization

→ TDMA (Time division Multiple Access)

2 CDMA

- Allocate different codes to different users.

- Orthogonal codes are used.

Let C_i is orthogonal codes

$$C_i \cdot C_i = N \quad (\text{No of Stations})$$

- Walsh table is used to generate orthogonal codes.

$$W_1 = [+1]$$

$$W_2 = \begin{bmatrix} W_1 & W_1 \\ W_1 & \overline{W_1} \end{bmatrix}$$

$$W_4 = \begin{bmatrix} W_2 & W_2 \\ W_2 & \overline{W_1} \end{bmatrix}$$

$$\begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix} \begin{matrix} \rightarrow C_1 \\ \rightarrow C_2 \\ \rightarrow C_3 \\ \rightarrow C_4 \end{matrix}$$

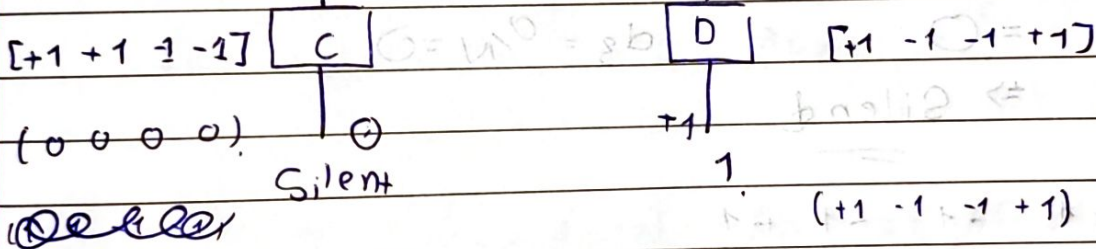
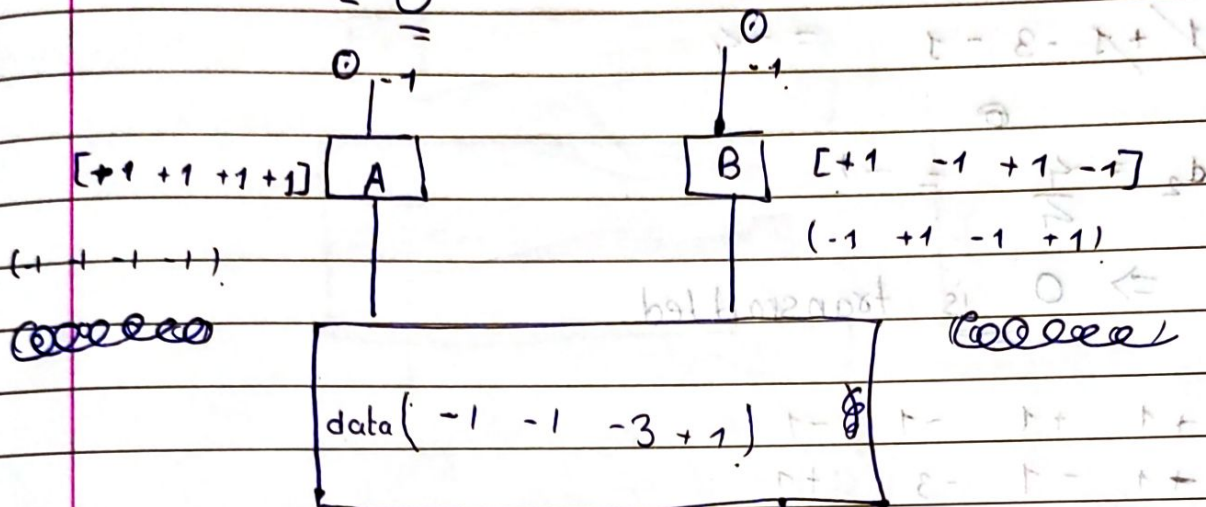
$$C_2 \cdot C_2 = \begin{bmatrix} +1 & -1 & +1 & -1 \end{bmatrix} \begin{bmatrix} +1 & -1 & +1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} +1 & +1 & +1 & +1 \end{bmatrix} = 4$$

$$C_1 \cdot C_2 = \begin{bmatrix} +1 & +1 & +1 & -1 \end{bmatrix} \begin{bmatrix} +1 & -1 & +1 & -1 \end{bmatrix}$$

$$= 1 - 1 + 1 - 1$$

$$= 0$$



$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

$$\text{Silent} \rightarrow 0$$

$$\text{data} = C_1 \cdot d_1 + C_2 \cdot d_2 + C_3 \cdot d_3 + C_4 \cdot d_4$$

$$C_1 (C_1 \cdot d_1) + C_1 (C_2 \cdot d_2) + C_1 (C_3 \cdot d_3) + C_1 (C_4 \cdot d_4)$$

$$Nd_1 + 0 + 0 + 0$$

d_1 is data transmitted by station 1.

$$d_1 = \frac{-1 - 1 - 3 + 1}{4} = \frac{-4}{4} = -1$$

$$\begin{array}{cccc} -1 & -1 & -3 & +1 \\ +1 & -1 & +1 & -1 \end{array}$$

$$-1 + 1 - 3 - 1 = -4$$

$$d_2 = \frac{-4}{4} = -1$$

$\Rightarrow 0$ is transmitted

$$\begin{array}{cccc} +1 & +1 & -1 & -1 \\ +1 & -1 & -3 & +1 \end{array}$$

$$-1 \quad -1 \quad +3 \quad -1$$

$$d_3 = 0/N = 0$$

$\Rightarrow$ Silence

$$+1 \quad -1 \quad -1 \quad +1$$

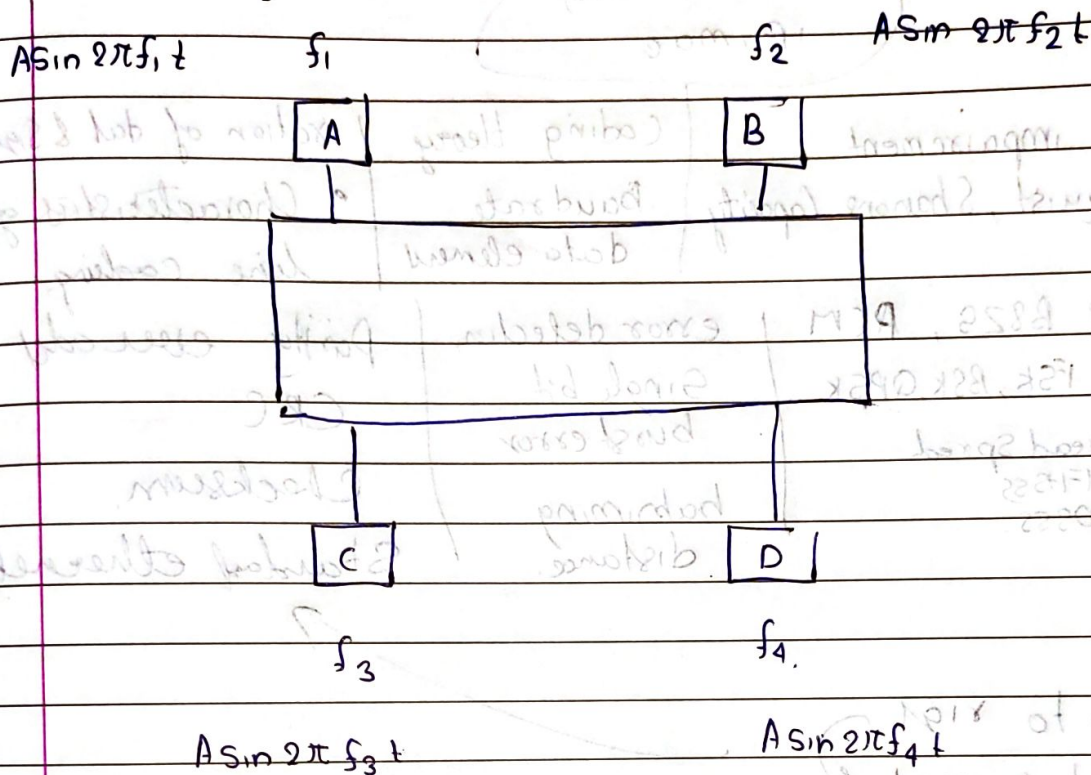
$$-1 \quad -1 \quad -3 \quad +1$$

$$-1 + 1 + 3 + 1 = 4$$

$$d_4 = \frac{4}{4} = 1$$

$\Rightarrow$ data transmitted 1

2. Frequency Division Multiple Access (FDMA)



$$S(t) = A \sin 2\pi f_1 t + A \sin 2\pi f_2 t + A \sin 2\pi f_3 t + A \sin 2\pi f_4 t$$

Filters — filter of f_1 will allow only f_1 and block others

3. TDMA (Time Division Multiple Access)

